

Seminar Paper on Resource Access Protocols

Raihan Uddin

April 2025

Abstract

In resource constrained environment of embedded electronic, resource allocation is critical consideration. For tasks to be completed efficiently and within time frame, protocols are to be followed to allocate resources sufficiently. We know that, in hard real time systems, deadline miss can pose life threatening consequences. The operating system of any real time system follows these protocols to invoke its kernel to allocate resources on-demand for the executable processes. The seminar paper is a discussion on how the resource protocols are followed inside the OS.

1 Introduction

”Starting a new process is a heavy job for the OS, which includes allocating memory, creating data structures, and coping code.”[1]

References

- [1] Jiacun Wang, *Real-Time Embedded Systems* (Quantitative Software Engineering Series). 111 River Street, Hoboken, NJ 07030, USA: John Wiley & Sons, Inc., 2017, ISBN: 9781119420705.